

Rajalakshmi Engineering College

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2024_28_III_OOPS Using Java Lab

2028_REC_OOPS using Java_Week 8_Q1

Attempt : 1
Total Mark : 10
Marks Obtained : 10

Section 1 : Coding

1. Problem Statement

Write a program to validate the email address and display suitable exceptions if there is any mistake.

Create 3 custom exception classes as below

DotException AtTheRateException DomainException

A typical email address should have a "." character, and a "@" character, and also the domain name should be valid. Valid domain names for practice be 'in', 'com', 'net', or 'biz'.

Display Invalid Dot usage, Invalid @ usage, or Invalid Domain message based on email id.

Get the email address from the user, validate the email by checking the

above-mentioned criteria, and print the validity status of the input email address.

Input Format

The first line of input contains the email to be validated.

Output Format

The output prints a Valid email address or an Invalid email address along with the suitable exception

If email ends with . or contains not exactly one . after @, it throws:

DotException: Invalid Dot usage

Invalid email address

If @ appears not exactly once, it throws:

AtTheRateException: Invalid @ usage

Invalid email address

If the part after the last dot is not among accepted domains:

DomainException: Invalid Domain

Invalid email address

If all conditions satisfied then print:

Valid email address

Refer to the sample input and output for format specifications.

Sample Test Case

Input: sample@gmail.com

Output: Valid email address

Answer

```
// You are using Java
import java.util.*;
class DotException extends Exception{
    DotException(String a){
        super(a);
    }
}
class AtTheRateException extends Exception{
    AtTheRateException(String b){
        super(b);
    }
}
class DomainException extends Exception{
    DomainException(String c){
        super(c);
    }
}
class Main{
    static int check(String a) throws
    DotException,AtTheRateException,DomainException{
        int f=1;
        if(a.endsWith(".") || a.startsWith(".") || a.contains("..") ){
            f=0;
            throw new DotException("Invalid Dot usage");
        }
        if(!(a.endsWith(".com")||a.endsWith(".in")||a.endsWith(".net")||
a.endsWith(".biz"))){
            f=0;
            throw new DomainException("Invalid Domain");
        }
        int c=0;
```

```

    for(char i:a.toCharArray()){
        if(i=='.') c++;
    }
    String []arr=a.split("@");
    if(arr.length==1 || c!=1){
        f=0;
        throw new DomainException("Invalid Domain");
    }
    if(arr.length!=2 || a.contains("@@")){
        f=0;
        throw new AtTheRateException("Invalid @ usage ");
    }
    return f==1?1:0;
}

public static void main(String args[]){
    Scanner sc=new Scanner(System.in);
    String a=sc.nextLine();
    try{
        int x= check(a);
        if(x==1){
            System.out.println("Valid email address");
        }
    }
    catch(DotException A){
        System.out.println("DotException: "+A.getMessage());
        System.out.println("Invalid email address");
    }
    catch(AtTheRateException B){
        System.out.println("AtTheRateException: "+B.getMessage());
        System.out.println("Invalid email address");
    }
    catch(DomainException C){
        System.out.println("DomainException: "+C.getMessage());
        System.out.print("Invalid email address");
    }
}
}

```

Status : Correct

Marks : 10/10